Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 12 GB of memory. Each experiment was repeated 3 times with different random seeds.

2 Results

Figure 1 shows the f1 score during training.

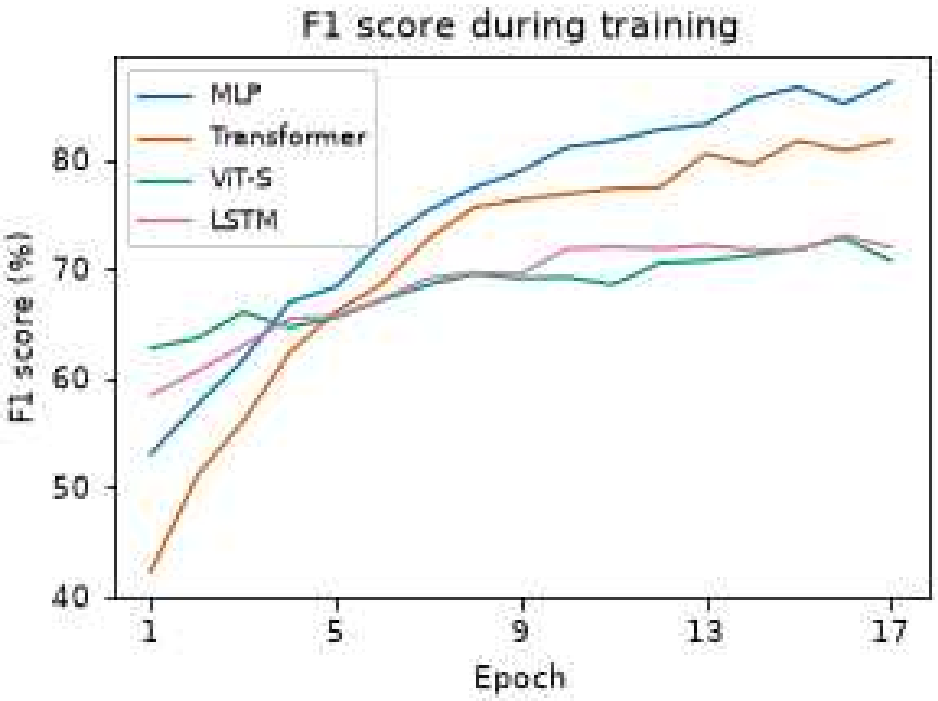


Figure 1: F1 score during training for the models in this experiment.

As shown in Figure 1, at epoch 15, LSTM achieves 70.7% F1 score. At epoch 9, Transformer is 7.2 points better than ViT-S.

Figure 2 shows the accuracy during training.

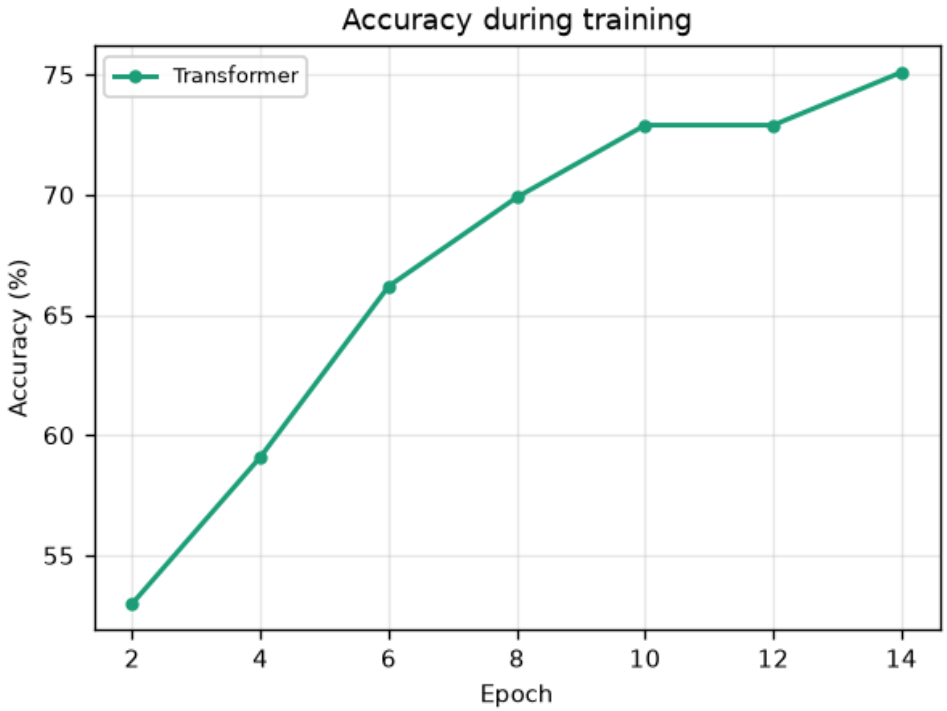


Figure 2: Accuracy during training for the models in this experiment.

Transformer reaches 60.7% accuracy at epoch 12. As shown in Figure 2, Transformer attains a accuracy of 75% at epoch 14.

Figure 3 shows the test accuracy of EfficientNet by dataset.

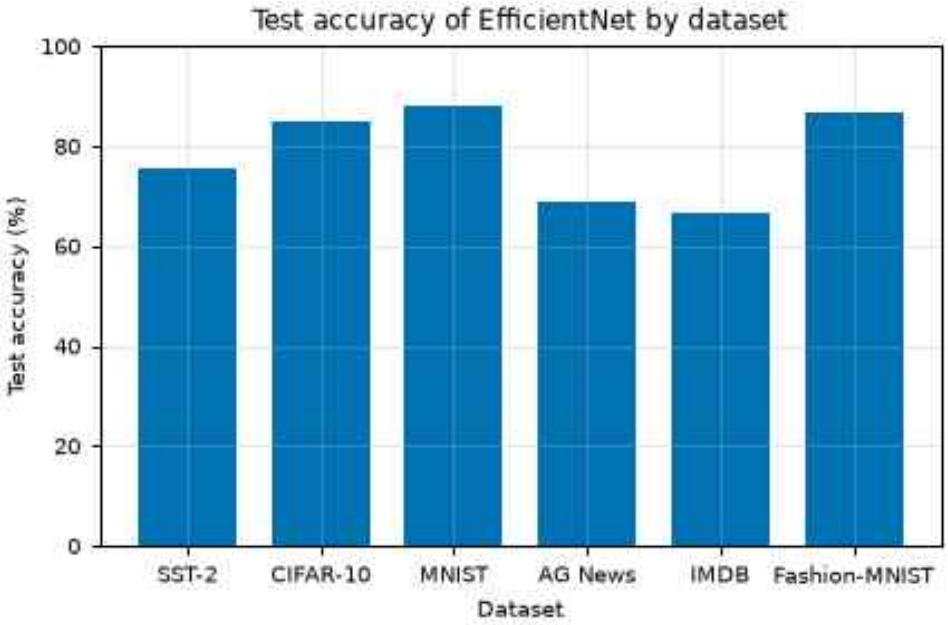


Figure 3: Test accuracy of EfficientNet by dataset for the models in this experiment.

EfficientNet scores 9 points higher on CIFAR-10 than on SST-2. On SST-2, EfficientNet obtains a test accuracy of 76% (Figure 3).

3 Discussion

Overall, the results are in line with what we expected from the literature.